

1			$m = \frac{T^2 k}{4\pi^2}$ $= \frac{3.2^2 (250)}{4\pi^2}$ $= 64.846 \text{ kg}$ $\frac{\Delta m}{m} = 2 \frac{\Delta T}{T} + \frac{\Delta k}{k}$ $\Delta m = \left(2 \frac{\Delta T}{T} + \frac{\Delta k}{k} \right) m$ $= \left(2 \frac{0.01}{3.20} + \frac{5}{250} \right) 64.846$ $= 1.7 \text{ kg}$ Thus, mass of the astronaut = $(65 \pm 2) \text{ kg}$	1	1	1
2	(a)	(i)	1			